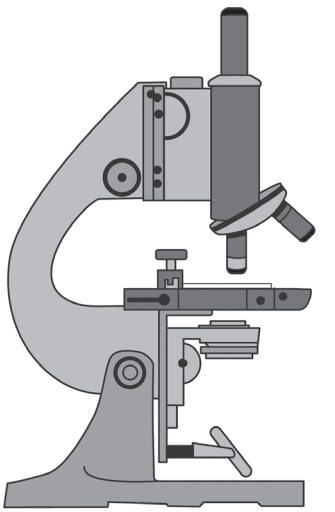
Single lens to compound microscope

The first microscope was used in the 1500s. Over time, microscope technology has steadily improved. As the technology improved so did knowledge about **microbes** – living things that are too small to see with the naked eye.



Compound microscope

- At first, **specimens** were magnified with single lenses. Some of the scientists involved at the time were Sacharias Jansen in around 1595, Robert Hooke in 1665 and Antonj van Leeuwenhoek in 1653.
- Hooke and Jansen later used two lenses in a row, sometimes inside a barrel. These were the first compound microscopes.
- Hooke was the first to try shining a light directly onto the specimen.
- Van Leeuwenhoek was first to see pond life protozoa, spirogyra and sperm of many animals.
- Hooke made very thin slices of cork that let light through, and saw little compartments, which he named **cells**.

☞ **1** Choose the correct word or phrase in each pair.

Microscopes

The lens of a [microscope/camera] lets only a tiny amount of [light/water] through. Light can be blocked by using too much of a specimen or a slice from a specimen that is too [wide/thick]. Robert Hooke cut thin slices of cork that let light pass through so he could see the cells. He also worked out a way to light up the specimen with a small reflecting [lamp/light globe].

Antonj van Leeuwenhoek used tiny lenses made by melting and pulling glass into bead shapes. He sometimes used [one/two] lenses – one in each hand – to get a [clearer/brighter] view. This was like a [compound/simple] microscope without the barrel. Improvements in lenses and microscopes over [tens/hundreds] of years show how the human endeavours of science and technology develop alongside each other.

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